

Views:

A **View** in MySQL is essentially a **virtual table**. It does not store physical data itself; instead, it stores a pre-defined SELECT query. When you query a view, MySQL runs the underlying query against the base tables and returns the result dynamically.

1. Why Use Views? (Benefits)

- **Security (Data Hiding):** You can restrict users from seeing sensitive columns (e.g., salaries, SSNs) by creating a view that only includes non-sensitive columns.
 - **Simplicity:** Instead of writing complex JOIN queries with WHERE, GROUP BY, and HAVING clauses over and over, you can save that query as a view and query it with a simple SELECT * FROM view_name.
 - **Consistency:** It creates a "single source of truth" for complex business logic. If a business rule changes, you only update the view, not every single query in your applications.
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2. Core Syntax Reference

Creating a View

SQL

```
CREATE VIEW view_name AS
```

```
SELECT column1, column2
```

```
FROM table_name
```

```
WHERE condition;
```

Querying a View

You treat a view exactly like a normal table:

SQL

```
SELECT * FROM view_name;
```

Modifying a View (Replacing)

If you need to change the query inside the view, use CREATE OR REPLACE:

SQL

```
CREATE OR REPLACE VIEW view_name AS
```

```
SELECT column1, column2, column3
```

```
FROM table_name
```

```
WHERE condition;
```

Dropping (Deleting) a View

This deletes the virtual table structure but **does not** delete data from the underlying base tables.

SQL

```
DROP VIEW IF EXISTS view_name;
```